

Ehsan Al Twal

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SKILLS

Software/Firmware: C, C++, Python, ROS 2, EtherCAT, Linux, RTOS, Git, Docker, Bash, Pytest, OpenCV, CNN, SLAM, DDS, NVIDIA Jetson, STM32, FOC, TypeScript, Doctest, J-Link, Cursor, MATLAB, Simulink, Java, C#, Assembly, VHDL, React, API

Electrical: Electrical Systems Design, PCB Design (Altium), BLDC Motor Control, ARM Microcontrollers (STM32, AX58400, Teensy 4.0), FPGAs (De1-SoC), Debugging & Soldering, SPI, I2C, CAN, RF, EtherCAT

EDUCATION

B.A.Sc. in Engineering Physics, University of British Columbia 2021 – Dec 2026
• Specialized in **Robot Software Engineering** Vancouver, Canada

PROFESSIONAL EXPERIENCE

Robot Software Engineer, Sanctuary AI Dec 2025 – Present
• **Implemented a custom fault publisher and recovery controller** to ensure safe fault recovery for bimanual dual UR robotic systems.
• **Architected "MockCAT," an EtherCAT slave simulator** that enables the software team to develop software and emulate full robotic hand networks without hardware.
• **Deployed an IgH EtherCAT master on an NVIDIA Jetson Orin** to control hydraulic robot hands, achieving stable 1 kHz control loops.
• **Benchmarked real-time control paths across computing architectures**, isolating DDS middleware latency to optimize the hand controller's network topology.

Firmware Engineer, Sanctuary AI May 2025 – Dec 2025
• **Developed robot's hydraulic power unit firmware on AX58400:** system monitoring, safety interlocks, fan control, ECAT communication, and dual-core IPC. Vancouver, Canada
• **Tripled team throughput** by building a local **EtherCAT** dev/test suite GUI, replacing shared Speedgoat resource, thus parallelizing development. Added a Pytest tool automating 75 PDO tests/sec.
• **Developed a full-stack tactile data pipeline** (firmware, EtherCAT Master and Slave, and ROS 2 bridge) to stream sensor data and enable live teleoperation haptics.

Electrical Engineer / Firmware Engineer, Sarcomere Dynamics May 2024 – Dec 2024
• Achieved a **3x product cost reduction** by designing a new electrical system and control firmware in C on STM32 for a new **robotic hand**. Vancouver, Canada
• **Extended product lifespan 1 to 5 years** by redesigning the electrical system with BLDC motors, designing a motor controller PCB, and developing an **FOC library**.
• **Reduced motor size by 30% and improved accuracy** by replacing Hall effect sensors with a magnetic sensor and current sensing.

PROJECTS

Autonomous Quadruped Guide Dog

- **Engineered an autonomous C++/ROS 2 navigation stack on Nvidia Jetson**, fusing FAST-LIO for 10 Hz real-time 3D odometry and mapping with a custom path-planning module to continuously generate optimal, collision-free routes with under **150 ms** latency.
- **Designed a custom hierarchical C++ motion controller** that fused a high-level Pure Pursuit state machine with a positional and angular PD heading controller, completely eliminating the need for basic PID fallbacks, to execute trajectory tracking with a positional accuracy of **2 cm**.
- **Deployed a real-time machine learning inference pipeline** that predicted the trajectories of dynamic obstacles **2 seconds** into the future to proactively trigger evasive path-planning maneuvers.